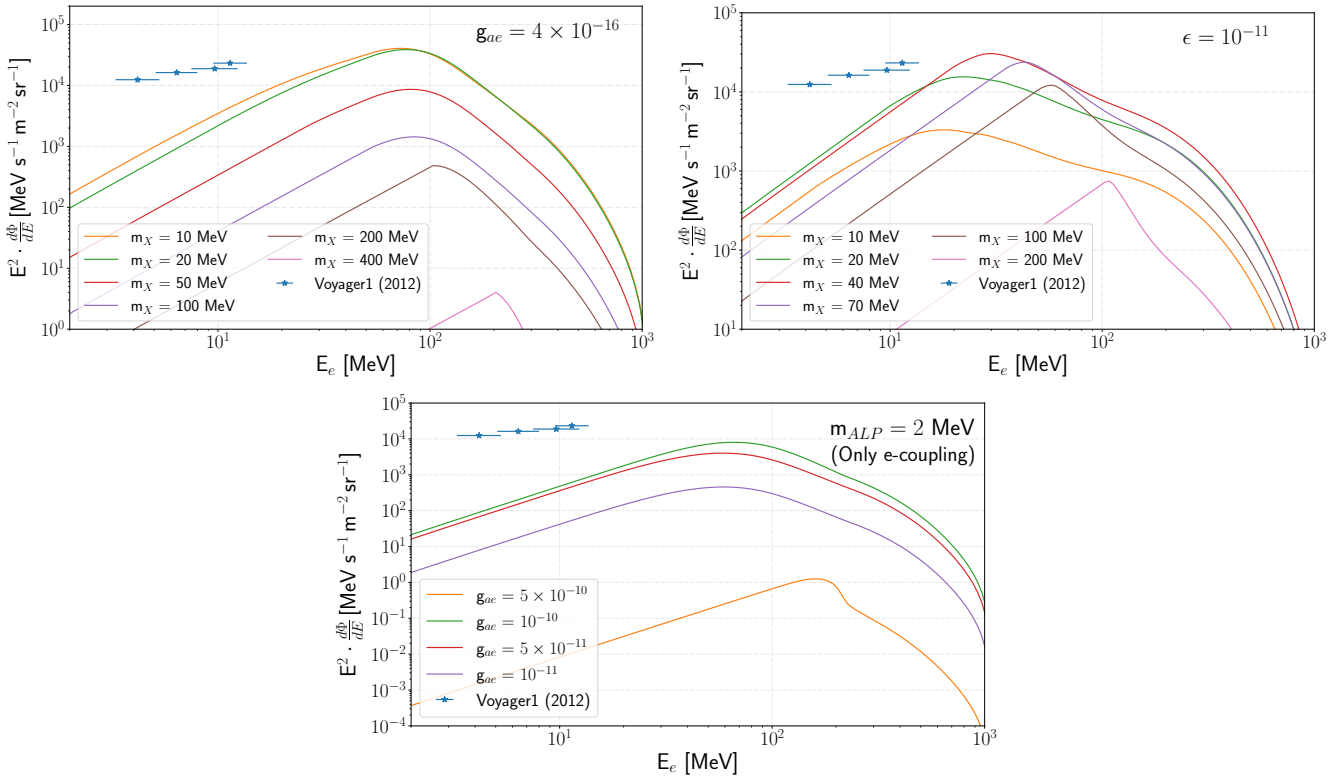

FIG. 4. Local diffuse $e^{\pm}$ flux predicted for different FIP models, compared to Voyager-1 measurements [98, 99]. The upper-left panel correspond to the case of ALPs coupling with both baryons and electrons (here we assume $g_{ap} = 2 \times 10^{-11}$), while upper-right panel refer to DPs. Different colors depict the case of different masses. The bottom panel illustrates the diffuse fluxes predicted for ALPs with a mass of 2 MeV coupling only to electrons.

in the DRAGON2 code. These tables are available upon request to the authors.

We show our estimations of the propagated local spectrum of $e^{\pm}$, in the MeV range, in Fig. 4, compared to Voyager-1 measurements [98, 99, 107]. The upper panels correspond to the case of ALPs (coupling with both baryons and electrons) in the left, and DPs in the right. For the case of ALPs, here we set the production by fixing $g_{ap} = 2 \times 10^{-11}$. These panels show that Voyager-1 observations cannot set very strong constraints for FIPs from SNe, given that their spectrum generally peaks around ~ 100 MeV, similarly to what we showed for the case of sterile neutrinos in Ref. [24]. In all these FIP models we observe that the expected diffuse flux of $e^{\pm}$ in the Galaxy starts to decrease significantly for masses above 100 MeV. As for the case of ALPs that only couple with electrons, the coupling needed to be constrained by Voyager-1 data (or secondary radiations) is so high ($g_{ae} > 10^{-9}$) that the ALPs cannot escape the SNe and they decay in their interior, therefore not producing any diffuse flux of $e^{\pm}$. This can be seen from the bottom panel of Fig. 4, where we show the local $e^{\pm}$ predicted for different coupling values, for a mass of 2 MeV. In this picture, the decrease in flux at couplings larger than $g_{ae} \gtrsim 10^{-10}$ is clearly evident between the green and orange curves with a strong flux suppression for $g_{ae} = 5 \times 10^{-10}$. We note that for larger masses, the fluxes at high g_{ae} are even more suppressed.

γ-ray signals

Given the limited data on low-energy CR electrons and positrons, previous works showed [26–28] that the coupling of FIPs with electrons and positrons can be better probed through the secondary emissions of the positrons injected. The main reason is that CR positrons are not very abundant, since they are mainly produced from interactions of high-energy protons with the ISM gas. One of the most promising emissions to constrain the FIP coupling is the 511 keV line emission produced when electrons in the ISM encounter the positrons injected by FIPs, once they have become thermal, and form the para-positronium bound state. However, recently, in Ref. [24] we showed that IA emission can be more effective to probe FIP production in the Galaxy. This emission stems from the direct annihilation of ISM electrons with relativistic positrons produced by FIPs and results in a continuum photon emission that peaks around a few tens of MeV, where backgrounds are expected to be very low.

We calculate the 511 keV diffuse line emission as we